JOHN HAROLD D. DOTON

Back-end Developer | Computer Science Student

harolddoton@gmail.com | 0936-029-6134 | San Rafael, Bulaon, San Fernando, Pampanga, Philippines

linkedin.com/in/harolddoton • github.com/Haruuowo • haroldportfolio.vercel.app

Technical Skills

- **Backend & Languages :** Python, JavaScript/TypeScript, Java, Go, Kotlin, Node.js
- **Databases & Cloud:** PostgreSQL, MySQL, Supabase, Firebase
- **DevOps & Architecture:** Docker, RESTful APIs, Git, GitHub
- **AI / ML:** Transformer fine-tuning (Whisper/Hugging Face), RAG, TTS pipelines, Transfer Learning, Scikit-learn
- **Frontend & Mobile:** React, Flutter, Dart, HTML5, CSS3

Experience

Internal Operations Officer — Angeles City, Pampanga
Google Developer Group (GDG) HAU — 2024 – Present

- Spearheaded technical workshops and coordinated community hackathons, facilitating hands-on learning for 100+ student developers in cloud and mobile technologies.

Projects

SmartSort (Intelligent File Organizer) — 2026
Python, SQLite, Pytest, PyWebView

- Architected the backend logic for a local file management system, implementing a YAML-based custom rule engine and content signature classification for automated sorting.
- Designed a relational database schema using SQLite to manage application state and log all file operations, ensuring data integrity and enabling a robust rollback/undo mechanism.
- Engineered concurrent real-time folder monitoring with exact-duplicate detection and EXIF-data parsing to optimize disk operations and system resource usage.
- Authored comprehensive backend unit tests using pytest to validate core algorithmic logic and database interactions.

Luna (Real-Time AI Streamer)
Python, REST APIs, WebSockets, RAG — 2026

- Developed the backend infrastructure and data pipelines for a real-time virtual streamer, integrating Twitch and TikTok Live APIs for concurrent event dispatching and chat processing.
- Implemented a Retrieval-Augmented Generation (RAG) vector system for state management, enabling contextual memory retrieval and continuous prompt construction.
- Orchestrated asynchronous LLM and Text-to-Speech (TTS) pipelines to ensure low-latency data flow and seamless real-time API responses.

Hands Free (Offline Voice-Control Architecture) — 2026
Lead Developer | Kotlin, On-Device Processing, System APIs

- Engineered a zero-dependency, local processing pipeline for an offline application, ensuring secure, on-device data handling without external network transmission.
- Designed an event-driven system utilizing an intent classifier and event dispatcher to trigger simulated system actions via the Android Accessibility Service API.
- Built and optimized an end-to-end data pipeline for automatic speech recognition (ASR), incorporating a translation normalization layer to parse and structure mixed-language inputs.

Certifications

- AWS Cloud Foundations | 2026 | Credly Badge
- Relational Database | 2024 | FreeCodeCamp Certificate
- JavaScript Essentials | 2024 | Credly Badge
- Responsive Web Design | 2024 | FreeCodeCamp Certificate

Education

Holy Angel University — Angeles City, Pampanga
Bachelor of Science in Computer Science, School of Computing — 2023 – Present

- Relevant Coursework: Data Structures & Algorithms, Machine Learning, Mobile Development, Software Engineering